IT Italiano · GB **English**



🔧 ichosynth — Build Manual

The real TŒRN firmware on hand-soldered hardware · 4 encoders · 3 buttons · OLED · 16×16 matrix

Difficulty Medium (SMD PSRAM) | runs TŒRN · SP_ | See also Usage Manual

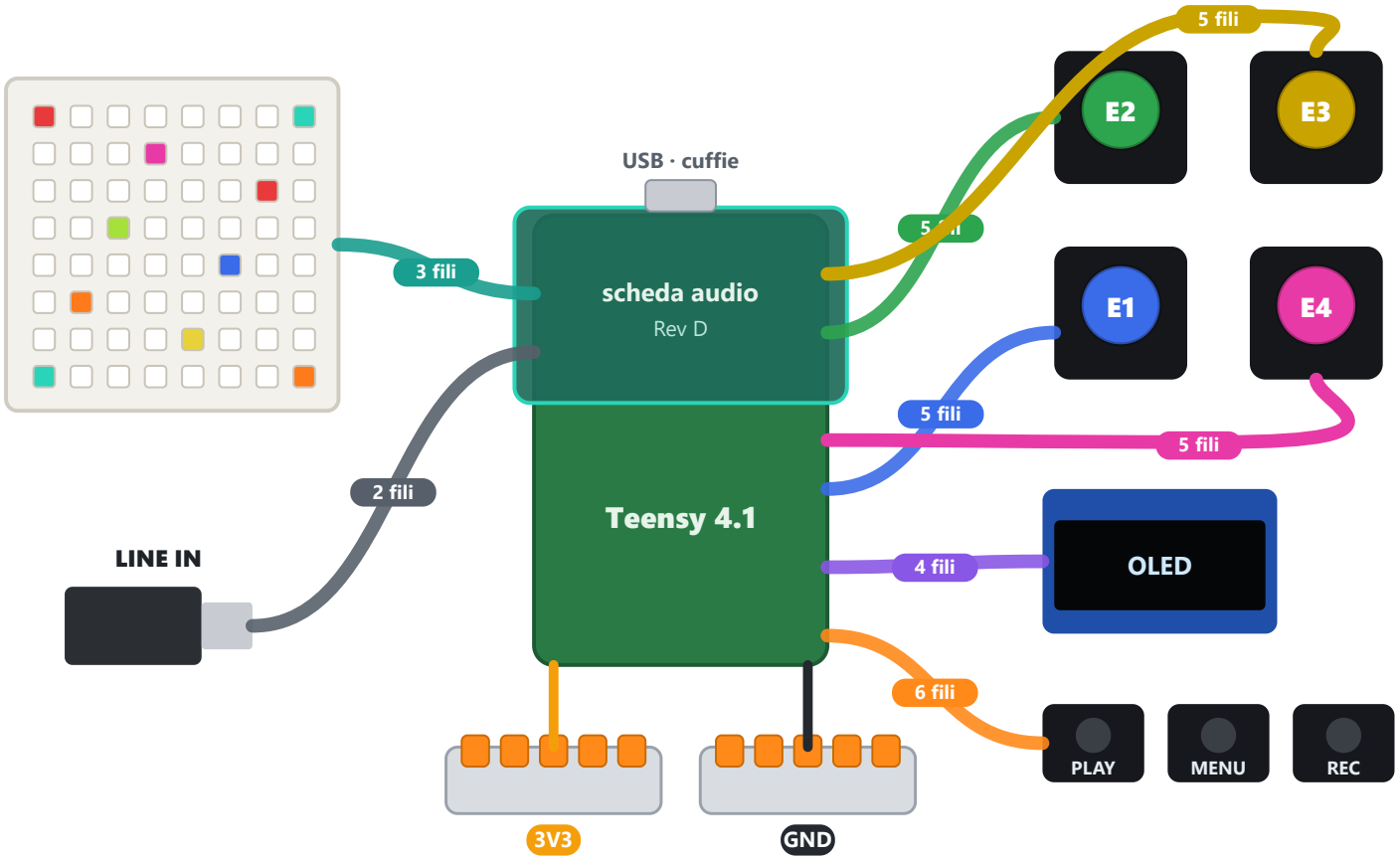
> 📄 **The graphic guides** (printable, almost no text — in Italian, but the pictures speak): **Wiring** — 7 plates, one wire at a time · **3D case assembly** — 18 pages · **Bench quick-start** — the checklist. This manual explains the why of each step and keeps the reference tables.

In short

ichosynth is a cheap, hand-soldered build of **TŒRN** (by SP_ / soundpauli, toern.live): an open-source sampler-sequencer on a **Teensy 4.1**. It runs the **real TŒRN firmware**; the expensive parts are replaced by cheap ones:

Original TŒRN	ichosynth
4 I²C RGB encoders (Duppa)	**4 KY-040 encoders** (turn + push)
3 capacitive pads	**3 tact switches** PLAY · MENU · REC
encoder ring colours	**0.96" OLED**: channel, mode, transport, BPM, volume, page
dedicated PCB	**soldered wires** (no PCB)

Everything else — DSP, sequencer, sampler — is the unchanged TŒRN firmware. A computer is only needed to load the firmware the first time; power comes from **USB (5 V)**.

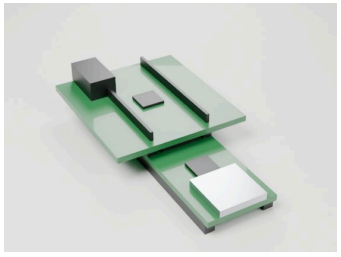
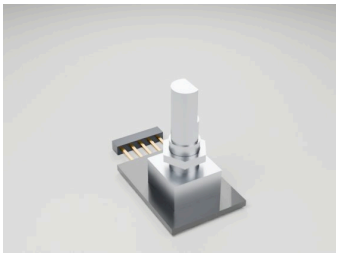
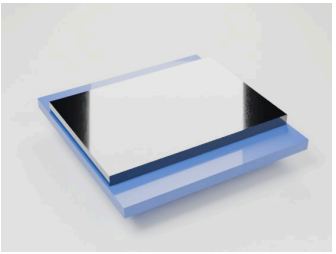
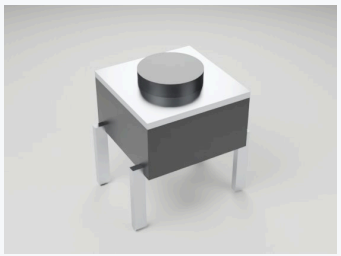
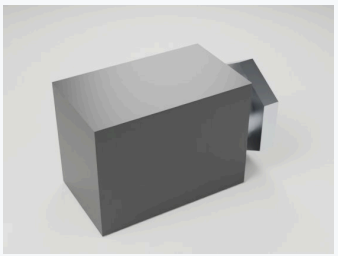
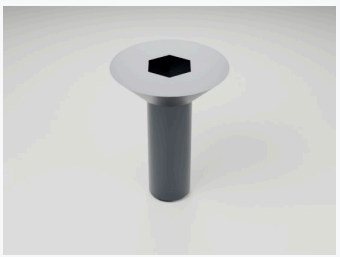


i fili + e GND di encoder, OLED, pulsanti e jack vanno ai morsetti

📑 Contents

1. What you need
2. Difficulty, tools, safety
3. Prepare the Teensy and the audio board
4. Wiring
5. 3D-printed case (optional)
6. Load the firmware
7. Prepare the micro SD
8. First power-on
9. If something is wrong
10. Pin summary

1 · What you need

 **Teensy 4.1 + Audio Rev D** ×1	**WS2812B 16×16 matrix** ×1	 **KY-040 encoder** ×4	 **SSD1306 0.96" OLED** ×1
 **6×6 tact switch** ×3	 **6.35 mm mono jack** ×1	 **M3×8 countersunk screws** ×8 (case)	 **Adhesive feet** ×4 (case)

Qty	Part	Notes
1	**Teensy 4.1**	the microcontroller
2	**8 MB PSRAM** chips (APS6404)	**both mandatory** (16 MB): TŒRN uses ~16.5 MB of external memory
1	**Teensy Audio Adaptor Rev D** (for Teensy 4.x)	SGTL5000 codec + 3.5 mm headphone jack
1	**WS2812B 16×16** LED matrix (256 LEDs)	rigid or flexible (160 mm)
4	**KY-040** encoder with push-button	E1 … E4
3	**6×6 mm** tact switches	PLAY · MENU · REC — **4.3 mm tall** if you use the case
1	**SSD1306 0.96" 128×64 I²C** OLED	replaces the encoders' colour rings
1	Panel-mount **6.35 mm mono TS** jack	LINE IN, for recording (the output is the audio board's headphone jack)
1	**Class 10** micro SD, ≤ 32 GB, FAT32	samples and patterns
1	**Data** micro-USB cable + 5 V ≥ 2 A supply	power and programming
1	Headphones with a 3.5 mm jack	no speakers on board
2	**Long male pin header** strips (≥ 15 mm, 14 pins)	to stack the audio board on the Teensy
2	*(recommended)* 5-pole **lever connectors** (e.g. WAGO 221-415)	3.3 V and GND bus bars, no soldering
as needed	AWG 26 wire in 6+ colours	~20 wires of 10-15 cm
1	*(optional)* **3D-printed case**	`case/` — 3 parts, 2 plates 250×250

ℹ️ **Samples**: the factory set in `sdcard/` is **public domain** (the *drum_sklad* collection), freely redistributable. You can add your own (see ch. 7).

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2 · Difficulty, tools, safety

Part	Difficulty	Notes
Wiring matrix, encoders, buttons, OLED	🟢 easy	soldering on large pins, one wire at a time
PSRAM (2 SMD chips under the Teensy)	🔴 hard	**buy the Teensy with the PSRAM already fitted**, or have it soldered

Tools: fine-tip soldering iron + solder (and flux), wire stripper, side cutters, tweezers, helping hands, **multimeter** (essential), isopropyl alcohol. An antistatic strap is recommended.

Four rules that save your parts:

1. **Never** connect or disconnect wires with USB plugged in.

2. **All grounds (GND) together.** Encoders and OLED at **3.3 V**, never 5 V: only the matrix gets 5 V.
3. After each group of joints, multimeter in continuity mode: **no short** between 5 V and GND or 3.3 V and GND.
4. Take your time: a "cold" joint (dull, ball-shaped) is the #1 cause of faults.

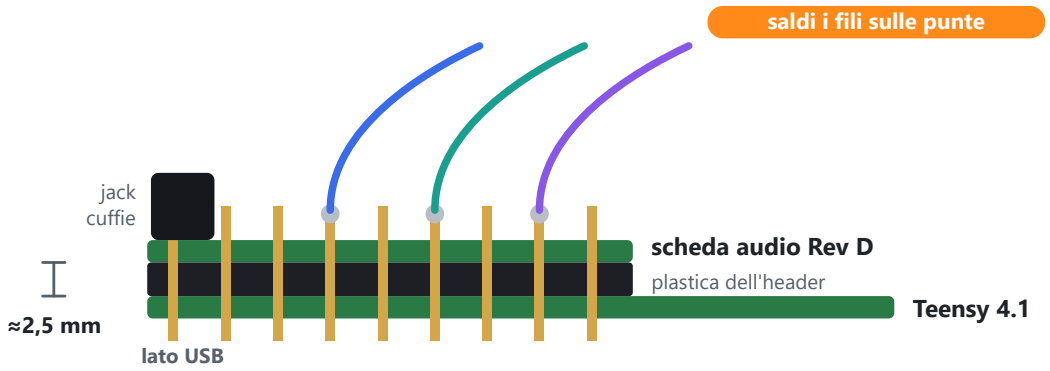
> ⏱️ Time: half a day if you already solder (PSRAM excluded).

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3 · Prepare the Teensy and the audio board

3.1 · PSRAM — if your Teensy doesn't have it yet, solder **both** chips on the pads underneath. Without both, the firmware won't start.

3.2 · Stack the audio board



1. Solder **2 strips of 14 long male headers** to the Teensy: the first 14 positions on each side, from the USB end down, plastic resting on the Teensy.
2. Slide the audio board on with its **headphone jack on the same side as the USB**, push it down until it **rests on the header plastic** (about 2.5 mm above the Teensy) and solder it.
3. The tips sticking out above the audio board **are the Teensy's pins**: the wires for the pins covered by the audio board are soldered there.

The audio signals (I^2S + I^2C) are then already connected. If you wire the audio board by hand instead:

Audio signal	Teensy pin		Audio signal	Teensy pin
MCLK	**23**		SDA (I^2C)	**18**
BCLK	**21**		SCL (I^2C)	**19**
LRCLK	**20**		3.3 V	**3.3V**
TX (data to codec)	**7**		GND	**GND**
RX (data from codec)	**8**			

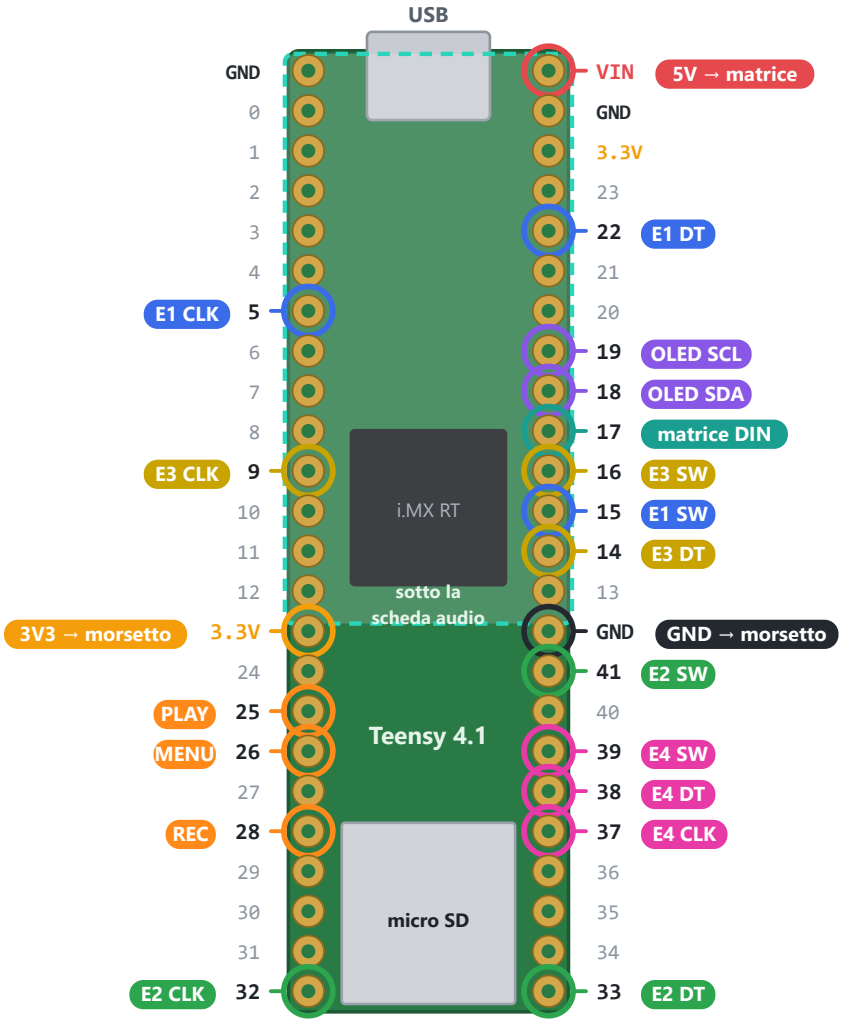
> ⚠️ **Don't fit the volume pot** on the audio board: its pad (A1) is pin 15, used by E1.

💾 The micro SD goes in the **Teensy 4.1's own slot** (at the bottom of the board), not the audio board's.

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4 · Wiring

🎨 Fully illustrated version for the bench: `docs/wiring/CABLAGGIO.pdf`. The pin numbers there and here come from the firmware drivers (`ICHOS_ENC_PINS` , `ICHOS_BTN_PINS` in `firmware/libraries/`).



The Teensy is seen **from the top with USB up**; coloured pins are the ones you use. The dashed area is covered by the audio board: those pins are soldered on the long header tips.

🧭 **Don't use pins 2 / 3 / 4** (internal to TŒRN) or **24** (LED strip removed in this build).

4.1 · Power: two lever connectors



From the Teensy	To the connector	Where
3.3V	**3V3** connector	left side, 15th pin from the top (outside the audio board)
GND	**GND** connector	right side, 15th pin from the top

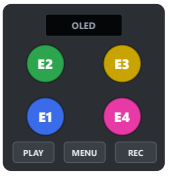
The **+** and **GND** of the 4 encoders and the OLED, and the GND of the buttons and the jack, go to them.

4.2 · KY-040 encoders (×4)



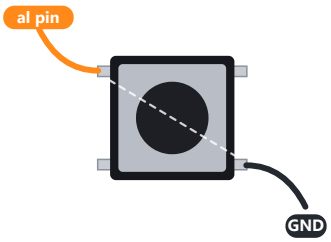
Encoder	CLK	DT	SW	+	GND
E1	**5**	**22**	**15**	3V3	GND
E2	**32**	**33**	**41**	3V3	GND
E3	**9**	**14**	**16**	3V3	GND
E4	**37**	**38**	**39**	3V3	GND

The labels **CLK DT SW + GND** are printed on the module: follow yours. If an encoder turns the wrong way, **swap CLK and DT**.



With the case the encoders sit 2×2: **E2 E3 on top, E1 E4 at the bottom** (E1 and E4 move the cursor in DRAW: the most used, nearest the hand). Without the case, lay them out as you like.

4.3 · PLAY · MENU · REC buttons



Button	Teensy pin	Other leg
PLAY	**25**	GND
MENU	**26**	GND
REC	**28**	GND

Use **two diagonal legs**: it always switches, whichever way you fit it. No resistors: the firmware uses the internal pull-ups.

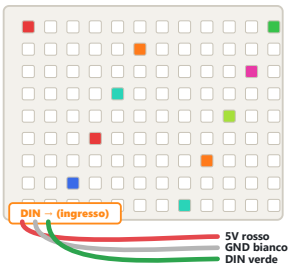
4.4 · OLED



OLED pin	Goes to
GND	GND connector
VCC	3V3 connector
SCL	pin **19**
SDA	pin **18**

It shares SDA/SCL with the audio board (different addresses: OLED 0x3C). ⚠️ On some modules **VCC and GND are swapped**: follow your board's labels.

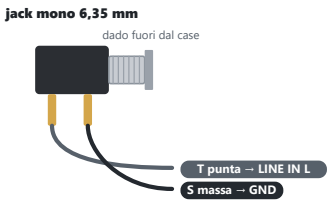
4.5 · LED matrix



Matrix wire	Goes to
5V (red)	pin **VIN** (right side, 1st from the top)
GND (white)	GND connector
DIN (green)	pin **17**

Use the connector where the arrow goes **in** (DIN), not DOUT. Wire colours vary between makers: the PCB label is what counts. The firmware keeps the brightness low, USB is enough.

4.6 · Audio: LINE IN and headphones



Jack	Connection
LINE IN 6.35 mm mono	tip → audio board **LINE IN L** pad, sleeve → GND
Headphones 3.5 mm	the audio board's own jack: no wires

LINE IN is mono (TŒRN samples the left channel) and is this build's default recording input. If the source hums, a 10 µF capacitor in series with the signal.

> 🎹 **MIDI (optional)**: TŒRN speaks DIN MIDI on Serial8 — **pin 34 = IN, 35 = OUT** — through a small optocoupler circuit (not included). The firmware does not handle MIDI over USB.

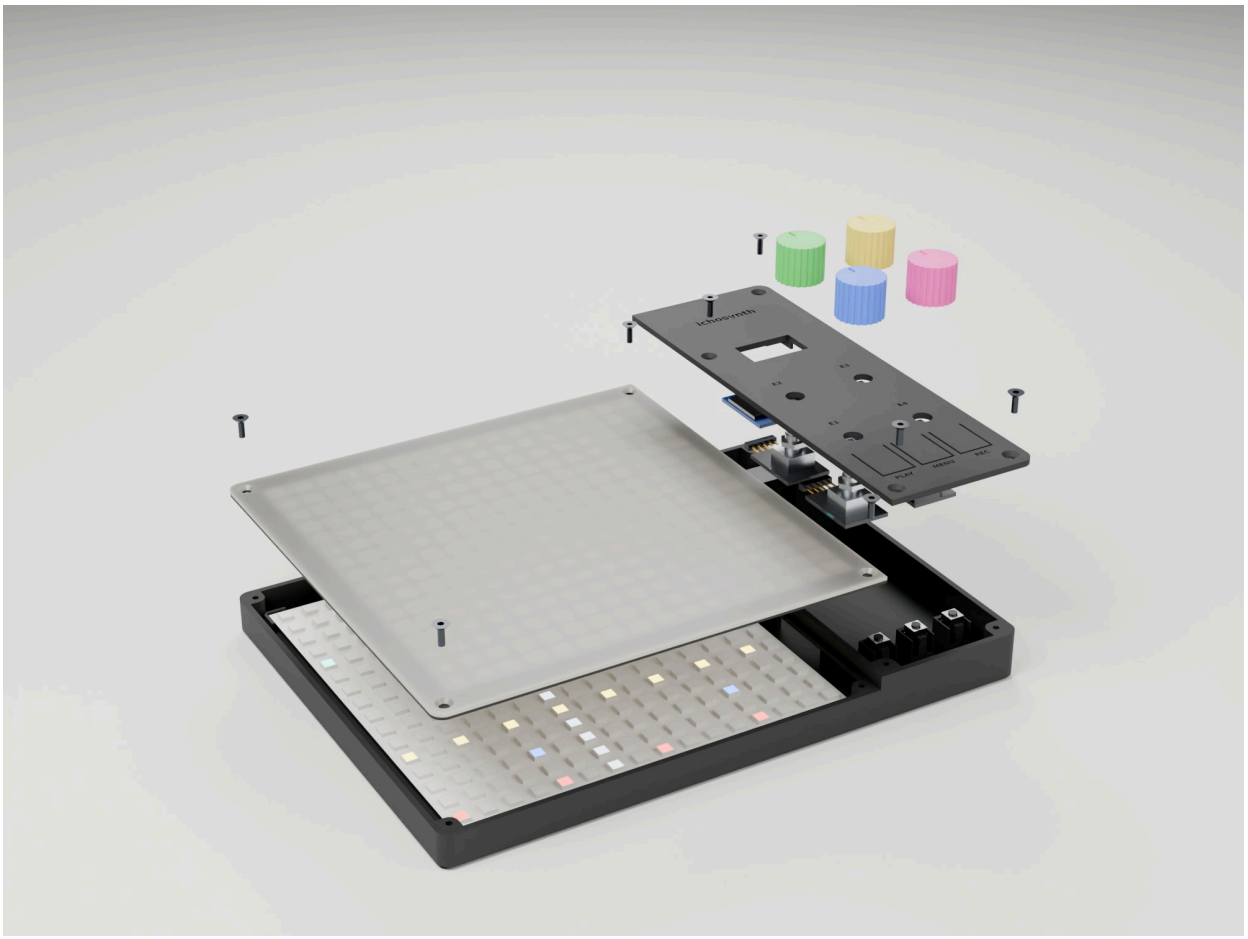
4.7 · Check before powering on

- [ ] every wire on the right number (compare with the map above);
- [ ] **only** the matrix's red wire goes to 5 V (VIN);

- [ ] multimeter: **no short** 5 V–GND and 3.3 V–GND;
- [ ] no solder bridges between neighbouring pins.

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5 · 3D-printed case (optional)



A stepped case, **3 parts on 2 plates** of the Anycubic Kobra S1, **no supports**: opaque tray and panel, translucent diffuser with the light-baffle grid built in (colour change at 1.4 mm). Headphones and USB exit at the back, LINE IN on the side; the buttons are flexible tabs cut into the panel.

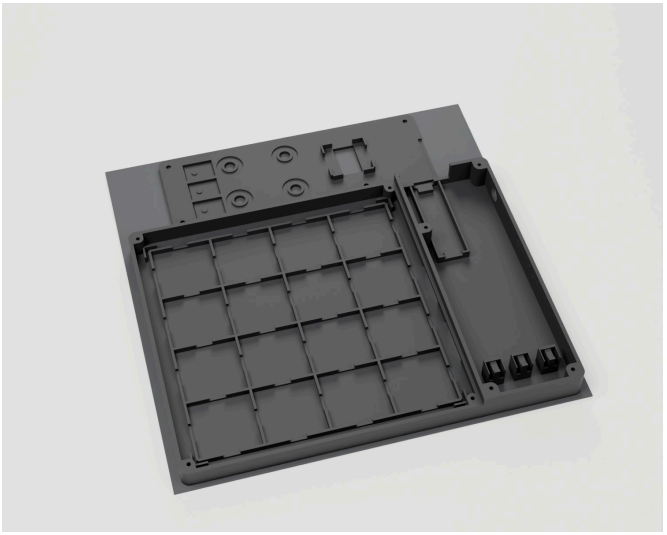


plate 1 · black

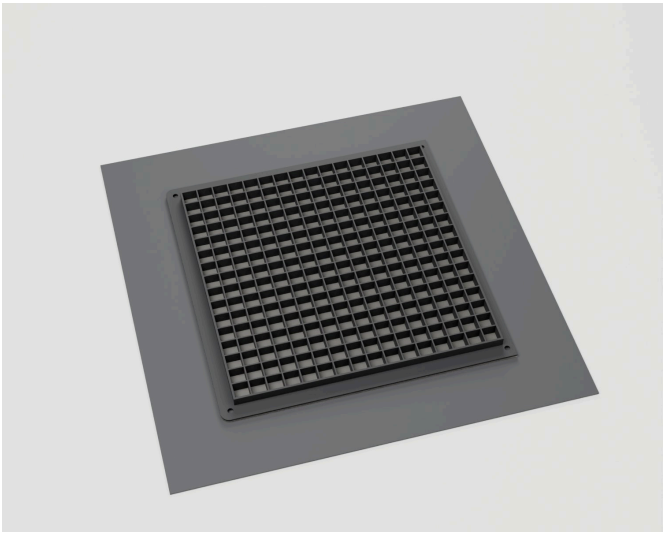


plate 2 · translucent

Ready-made **Anycubic Slicer Next** projects (Kobra S1 Combo + ACE Pro, slot 1 black, slot 2 translucent, colour change already set): `case/ichosynth_case_S1Combo.3mf` (all PLA) and `case/ichosynth_case_S1Combo_PETG.3mf` (all PETG). **Never mix PLA and PETG in one part**: they don't bond to each other.

👉 Step-by-step guide: **`case/MONTAGGIO.pdf`** · printing and parameters: `case/README.md`. With the case: encoders 2×2, **4.3 mm** switches, soldered wires (no Dupont), LINE IN jack ≤ 22 × 16 mm behind the wall.

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6 · Load the firmware

The easy way — one-click flasher (Windows):

1. connect the Teensy over USB;
2. open **`tools/flasher/flasher.exe`** (first run, SmartScreen: *More info → Run anyway*);
3. press ⚡ **Flash ichosynth** and, when asked, the small **PROGRAM** button on the Teensy;
4. wait for "*Firmware written and Teensy rebooted*".

The flasher already contains the compiled firmware (`tools/flasher/ichosynth.hex`) and checks the file is intact before writing it.

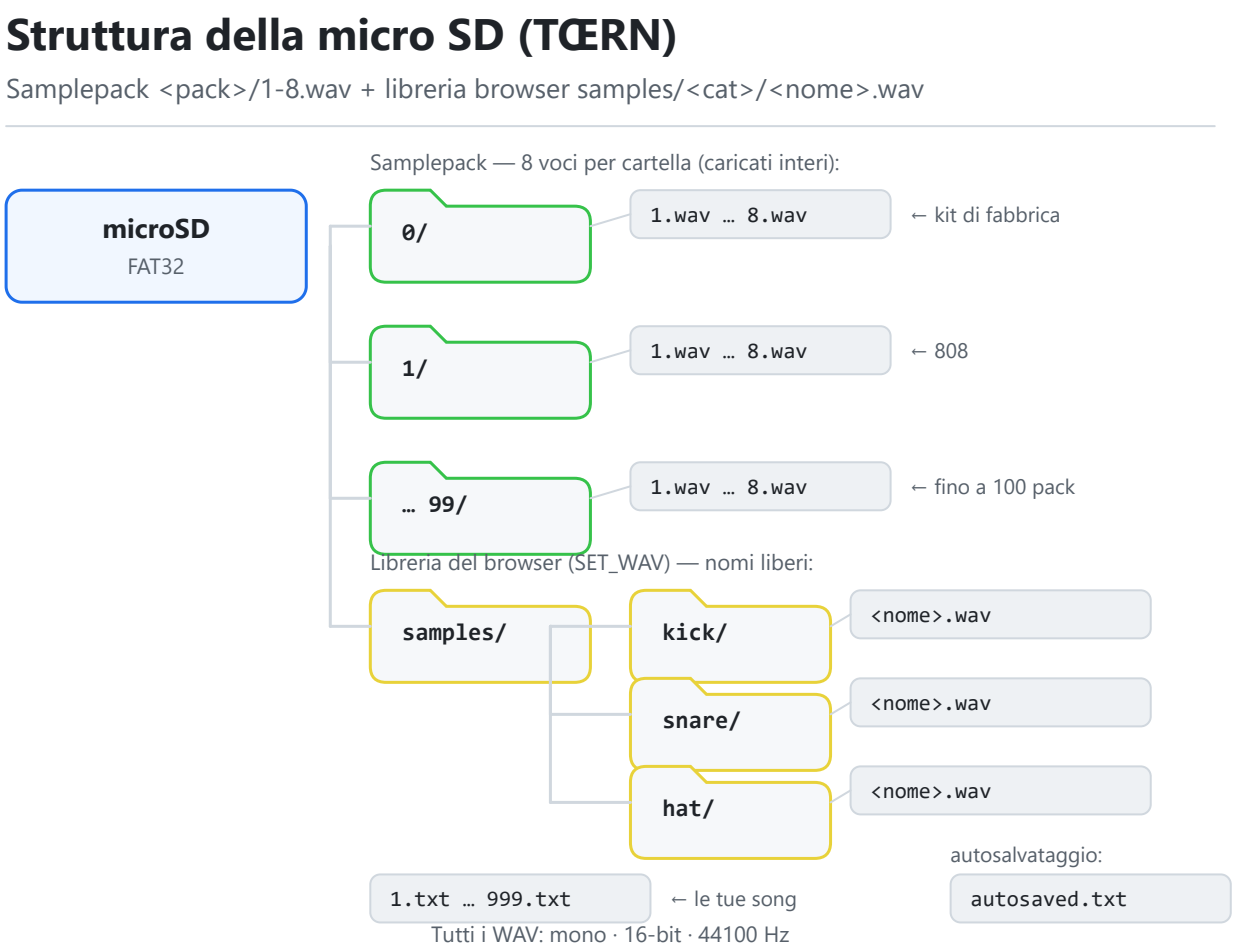
Building from source:

```
python firmware/build_toern.py            # -> firmware/build/toern.hex
python firmware/build_toern.py --flash    # build and flash (teensy_loader_cli)
```

Needs `arduino-cli` with the `teensy:avr` core (≥ 1.61.0). The script takes the TŒRN sources at the verified version, adapts the pins to our parts, uses the Teensy core's FastLED and compiles at `-O1` (at `-O2` the compiler crashes on TŒRN's ~23,000-line file). Details and patches: `firmware/README.md`.

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7 · Prepare the micro SD



/1-8.wav and browser library samples//.wav" width="620">

- **Samplepacks** — folders `0` … `99` at the root, each with **8 voices** `1.wav` … `8.wav` (one per channel). Pack 0 is the factory kit.
- **Browser library** — `samples/<category>/<name>.wav`, browsed voice by voice in **SET_WAV** (up to 256 files per folder, names up to 60 characters).
- **Format: mono WAV, 16-bit, 44100 Hz.**

The quickest way: **copy the contents of** `sdcard/` **to the root** of a FAT32 SD card (6 packs + library ready to go).

Adding your own samples — `tools/wavmaker/wavmaker.exe` (or `python tools/wavmaker/wavmaker_gui.py`):

1. pick the **mode**: *Samplepack* (8 voices → `<pack>/1-8.wav`, order with ↑ ↓) or *Library*;
2. **Add files/folder**: green rows are already in the right format;
3. **Scan** the SD (finds the next free pack) or write to a local folder;
4. **Convert**. The originals stay unless you choose to delete them.

wavmaker only formats **removable SD cards** and asks for confirmation showing disk and partitions. To regenerate the factory set: `python tools/wavmaker/build_samples.py`.

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8 · First power-on

1. SD in the Teensy slot, headphones in, USB.
2. The matrix shows the **logo**; the OLED shows **CH / MODE / BPM / VOL / PAGE**.
3. Red **noSD** icon? Power off, check the SD (FAT32, Teensy slot), power on again.
4. Turn the encoders: cursor and values respond. Press **PLAY**: the sequence starts.
5. Plug a source into LINE IN and use **REC** to record it into a channel (see the Usage Manual).

> 🎮 To play it: **Usage Manual** and the control map `docs/reference/MAPPA_CONTROLLI.md`.

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9 · If something is wrong

Symptom	Fix
🕯️ The PC doesn't see it	**data** USB cable (not charge-only); 5 V–GND short; header joints
🔁 It reboots by itself	PSRAM missing or badly soldered (2 chips needed); ≥ 2 A supply
💡 LEDs off or wrong colours	DIN not on pin 17; output connector instead of input; no common GND
↩️ An encoder turns the wrong way	swap that encoder's **CLK and DT**
⏩ One click moves two steps	set `ICHOS_ENC_COUNTS_PER_DETENT` to 2 in the driver and rebuild
🔌 Encoder or button does nothing	wrong pin: check the map; a button goes pin → GND
💾 noSD icon	SD in the **Teensy's** slot, FAT32, ≤ 32 GB
🎧 No sound	audio board badly stacked; volume at 0; use the audio board's headphone jack
🚫 A channel is silent	WAV not mono/16-bit/44.1 kHz or wrong path: use wavmaker
📟 OLED stays black	VCC/GND swapped; SDA/SCL swapped; address 0x3D
❌ The build fails	`arduino-cli` + `teensy:avr` core installed? run `python firmware/build_toern.py`

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10 · Pin summary

```
LED matrix    DIN 17 · 5V from VIN · GND
Encoder E1    CLK 5   DT 22  SW 15        Audio (stacked)   MCLK 23 · BCLK 21 · LRCLK 20
Encoder E2    CLK 32  DT 33  SW 41                          TX 7 · RX 8 · SDA 18 · SCL 19
Encoder E3    CLK 9   DT 14  SW 16        OLED              SDA 18 · SCL 19 · 3V3 · GND (0x3C)
Encoder E4    CLK 37  DT 38  SW 39        LINE IN           tip → LINE IN L · sleeve → GND
Buttons       PLAY 25 · MENU 26 · REC 28  (pin → GND)        DIN MIDI (opt.)   IN 34 · OUT 35
Avoid         2 / 3 / 4 (TEENSY internal) · 24 · don't fit the volume pot (A1 = pin 15)
Power         matrix 5 V (VIN) · encoders, OLED, audio 3.3 V · all GNDs together
```

Happy building! 🛠️ When it plays, move on to the **Usage Manual**.

*ichosynth is a cheap, hand-soldered build of **TŒRN** by SP_ (soundpauli, https://toern.live) · open-source firmware on a Teensy 4.1.*